# STAFF REVIEW · H446 004 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q5

## Do Now — source answers

1. Real / float: a number with a fractional part.

2. 00111100 = 32 + 16 + 8 + 4 = 60.

3. +26 = 00011010; invert → 11100101; add 1 → 11100110 .

4. 01100000 = 96. The leading 1 is discarded; 176 × 2 = 352 exceeds the byte’s maximum 255.

## I Do — source worked steps

1. Write 6.5 in binary — CODE:
6.5 = $110.1_2$ — The whole-number part is 110 ; the 0.5 fractional part is .1 .

2. Move the binary point — CODE:
$110.1_2$
= $0.1101_2 \times 2^3$ — The binary point moves three places left, so the exponent is 3 .

3. Write the two fields — CODE:
Mantissa: 01101000
Exponent: 0011

Stored: 01101000 0011 — Check: 0.8125 × 8 = 6.5 .

## We Do — guided CHECK

1. Build the answer
−2 in four-bit two’s complement is 1110 .

2. Build the answer
Check: −0.8125 × 8 = −6.5.

3. Build the answer

4. Build the answer
$-1 \times 2^{-1} = -0.5$.

5. Build the answer

## Page 3 — independent skill check, teacher-review model

1. -0.375 = -0.11 binary = -0.75 x 2^-1. Mantissa 10100000; exponent 1111. Normalised sign bits are 10.

## You Do — source mark schemes

1. CODE:
10.5 = $1010.1_2$ = $0.10101_2 \times 2^4$ → 01010100 0100

$0.375 = 0.011_2 = 0.11_2 \times 2^{-1} \rightarrow 01100000\ 1111$
Each: correct binary magnitude/scaling working (1); both final fields (1).

2. CODE:
−10.5: invert 01010100 → 10101011; +1 → 10101100
Final: 10101100 0100
−0.375: invert 01100000 → 10011111; +1 → 10100000
Final: 10100000 1111
Each: positive binary magnitude/scaling (1), correct negative mantissa (1), complete normalised fields (1).

3. CODE:
00001011 0111
00010110 0110
00101100 0101
01011000 0100
Three left shifts (1); exponent reduced by 3 (1); mantissa 01011000 (1); exponent 0100 (1). Check: original 11/128 × 128 = 11; final 88/128 × 16 = 11.

4. The mantissa was doubled but the scale stayed the same, so the value doubled (1). Reduce exponent 3 to 2 / 0010 (1). Correct fields: 01010000 0010 (1). Normalisation removes redundant sign bits to make full use of the available significant bits (1).

5. Mantissa 10000000 (1); exponent 1111 (1). $-1 \times 2^{-1} = -0.5$. 11000000 0000 has the correct value but is not normalised.

## Exit Ticket — source answer

1. True. More mantissa bits give greater precision to the represented value.